The dimension of silent subspaces in finite-sampling truncations of the Weil quadratic form

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Abstract

We identify and determine the dimension of the *silent subspace* of a finite-sampling truncation: the space of test vectors whose in-band contribution to the truncated Weil quadratic form vanishes identically. Our main result is exact and unconditional: if L collects N Fourier-sampling conditions together with one pole condition on a d-dimensional space of even functions, then

$$\dim\ker L \;=\; d - \operatorname{rank} L \;\geq\; \max(0,\, d - N - 1),$$

with equality in the lower bound precisely when the conditions are independent. Two independent prolate computations, carried out at 65-digit working precision and at two different Legendre cutoffs, confirm a two-dimensional joint kernel in both test cells, with agreement to about 22 relative digits. We further give a min–max eigenvalue bound over the silent subspace and, under positivity assumptions, a quantitative bound $\|P_{H^\perp} v\|^2 \leq \lambda/b$ on the out-of-kernel component of an eigenvector, together with an explicit 2×2 example showing that this bound cannot be strengthened to exact containment. The dimension theorem explains, from first principles, the empirical design rule $d = N + 3$ used in earlier truncation experiments.

1 Introduction

Finite truncations of Weil's quadratic form are the practical instrument by which positivity criteria attached to the Riemann hypothesis are explored numerically. In such a truncation one fixes a finite-dimensional space V of test functions, samples a transform at N prescribed ordinates, and imposes a normalisation (here: a pole condition). A recurring empirical observation is that certain test vectors contribute *exactly zero* to the in-band part of the form, and that the number of such vectors is governed by the excess of the dimension d over the number of conditions.

This note makes that observation precise. The silent subspace is the joint kernel of the sampling and pole conditions, its dimension is exactly $d - \operatorname{rank} L$, and the resulting lower bound $\max(0, d-N-1)$ accounts for the design rule $d = N+3$ that had previously been adopted without justification: it is the smallest dimension guaranteeing a two-dimensional silent subspace.

Section 2 fixes the setting and the basis convention. Section 3 proves the dimension theorem. Section 4 gives the spectral consequences that do follow, and delimits by counterexample one that does not. Section 5 reports the numerical confirmation. Section 6 states the scope of the results.

2 Setting

Let $h > 0$ and let V be a d-dimensional inner-product space of real-valued even functions on $[-h, h]$. Define

$$F_\eta(\gamma) = \int_{-h}^{h} \eta(u)\cos(\gamma u)\,du, \qquad \ell_\pm(\eta) = \int_{-h}^{h} \eta(u)e^{\pm u/2}\,du.$$

Remark 1. By evenness and the substitution $u \mapsto -u$ one has $\ell_+ = \ell_-$ identically; no numerical test of parallelism is needed. We write ℓ for the common functional.

Remark 2 (Basis convention). In the prolate realisation the modes $\psi_0, \psi_1, \ldots$ alternate in parity. The even space of dimension d is therefore $V_d^+ = \operatorname{span}(\psi_0, \psi_2, \ldots, \psi_{2d-2})$; taking the first d modes and then restricting to even functions would yield dimension $\lceil d/2 \rceil$ instead. All statements below refer to the former convention.

3 The dimension theorem

Fix distinct real sampling points $\gamma_1, \ldots, \gamma_N$ and define $E : V \to \mathbb{R}^N$, $L : V \to \mathbb{R}^{N+1}$ by

$$E\eta = \big(F_\eta(\gamma_1), \ldots, F_\eta(\gamma_N)\big), \qquad L\eta = (E\eta,\ \ell(\eta)).$$

Write $W := \ker L$ for the *silent subspace*.

Theorem 3. $\dim W \;=\; d - \operatorname{rank} L \;\geq\; \max(0,\, d - N - 1)$.

Proof. The rank–nullity theorem gives $\dim\ker L = d - \operatorname{rank} L$, and $\operatorname{rank} L \leq \min(d, N+1)$ since L maps into $\mathbb{R}^{N+1}$. □

Corollary 4 (The design rule $d = N + 3$)**.** *For $d = N + 3$ one has $\dim W \geq 2$, and $\dim W = 2$ exactly when $\operatorname{rank} L = N + 1$, i.e. when the N sampling conditions and the pole condition are independent. Thus $d = N + 3$ is the smallest truncation dimension guaranteeing a two-dimensional silent subspace.*

Repeating an evaluation point does not create an independent row; multiplicity of a zero changes weights but not the rank count.

Proposition 5 (Silence)**.** *Let $a_1, \ldots, a_N > 0$ and set $Q_{\text{in}}(\eta) = \sum_{j=1}^{N} a_j |F_\eta(\gamma_j)|^2$. Then Q_{in} vanishes identically on W. If moreover a decomposition $Q = Q_{\text{in}} + Q_{\text{out}}$ holds, then $Q|_W = Q_{\text{out}}|_W$.*

The identification of such a decomposition with the arithmetic Weil form is a separate hypothesis, not used in Theorem 3.

Remark 6. The value $d = N + 1$ is the dimension at which the guaranteed joint kernel may first vanish. This is a statement about $\dim W$, not about the size of the eigenvalues: in the data of Section 5 the largest jump in magnitude occurs at the transition $d = 31 \to 30$ (about 3.598 orders) rather than at $d = 32 \to 31$ (about 2.299 orders). The two phenomena should not be conflated.

4 Spectral consequences

Let $K = B + C$ be self-adjoint on a finite-dimensional space with $B \geq 0$ and $C \geq 0$, let $H = \ker B$ and $k = \dim H > 0$. Ordering eigenvalues increasingly from λ_0, the min–max principle with trial space H gives

$$\lambda_{k-1}(K) \;\leq\; \max_{w \in H,\ \|w\|=1} \langle w, Cw \rangle. \tag{1}$$

If a pole constraint is present, compress onto $\ker \ell$ first and apply the same statement.

Proposition 7 (Out-of-kernel bound)**.** *Suppose in addition $B \geq bI$ on $H^{\perp}$ with $b > 0$. Then for every eigenvector $Kv = \lambda v$ with $\|v\| = 1$,*

$$\|P_{H^{\perp}} v\|^2 \;\leq\; \lambda / b.$$

Proof. $\lambda = \langle v, Bv\rangle + \langle v, Cv\rangle \geq \langle v, Bv\rangle \geq b\|P_{H^{\perp}} v\|^2$. □

Thus an eigenvector is close to the silent subspace when λ/b is small. The following example shows that closeness cannot be upgraded to containment by (1) alone.

Proposition 8 (Sharpness)**.** *Let*

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad K = B + C = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

Both $B, C \geq 0$ and $H = \ker B = \operatorname{span}\{(0,1)\}$, yet

$$\lambda_{\min} = \tfrac{3-\sqrt{5}}{2} \approx 0.38196601125, \qquad v_{\min}^{\top} B v_{\min} = \tfrac{1-1/\sqrt{5}}{2} \approx 0.27639320225 > 0,$$

so $v_{\min} \notin H$.

Exact silence of the smallest eigenvectors therefore requires additional structure, such as invariance of H under K together with a spectral gap; Proposition 7 is what holds in general.

5 Numerical confirmation

We take $q = 13$, $h = \log(13)/2$ and $c = \Omega h$. The first d even prolate modes are obtained from a finite approximation of the prolate differential operator in normalised even Legendre polynomials. Sampling transforms are evaluated with spherical Bessel functions and the pole row with modified Bessel functions. All computations use 65-decimal working precision, at two separate Legendre cutoffs (88 and 112 even terms). The numerical rank threshold is 10^{-35} times the largest singular value.

Cell	Ω	N	even modes d	$\operatorname{rank} E$	$\operatorname{rank} L$	$\dim W$
A	75.5524	18	21	18	19	2
B	103.7255	30	33	30	31	2

Both cells realise the equality case of Theorem 3: the conditions are independent, $\operatorname{rank} L = N + 1$, and $\dim W = d - N - 1 = 2$. The ranks agree at both Legendre cutoffs. The smallest singular value of the joint matrix is about $1.4148646 \times 10^{-12}$ in cell A and $7.6443343 \times 10^{-17}$ in cell B, the latter agreeing to about 22 relative digits across the two cutoffs. Residual norms of the constructed unit kernel vectors are about $2\text{–}3 \times 10^{-67}$.

In the alternative reading that takes all parities among the first d modes, the even dimensions become 11 and 17, and the numerical joint kernel is trivial in both cells; this is consistent with Theorem 3, since there $d < N + 1$. The basis convention is therefore essential, and we use the one fixed in Section 2.

The Ω values are the rounded ordinates; in cell B the next computed ordinate is $103.725538040478\ldots$ Ordinates were obtained from a standard high-precision implementation.

6 Scope

The results proved here are Theorem 3, Proposition 5, the min–max bound (1), Proposition 7 and the sharpness example of Proposition 8; the numerical section confirms the equality case in two independent cells. Three points delimit the scope. First, for a zero $\rho = \beta + i\gamma$ the natural spectral parameter is $z_\rho = (\rho - \frac{1}{2})/i = \gamma - i(\beta - \frac{1}{2})$, which is real only under the Riemann hypothesis; accordingly we make no unconditional sum-of-squares claim for the arithmetic Weil form. Second, the numerical residuals above are algebraic residuals of the finite approximation and are not certified bounds on the prolate approximation error. Third, the dimension and spectral bounds are applications of standard linear algebra to this particular pair of conditions; any claim of novelty concerns the identification of the silent subspace in this setting, not the linear algebra itself. Production code and full-precision parameters are retained by the author.

Changes in this version

This version corrects four points of the first release: the abstract no longer reports three confirmed predictions, since the magnitude prediction was not confirmed (Remark 6); the largest magnitude jump is located at $d = 31 \to 30$ rather than at $d = N+1$; the claim that the smallest eigenvectors lie in the silent subspace is replaced by Proposition 7, with Proposition 8 showing why; and the unconditional positivity claim is withdrawn, as recorded in Section 6.

References

- Groskin, operator and explicit-formula dictionary: `https://arxiv.org/abs/2607.02828`
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